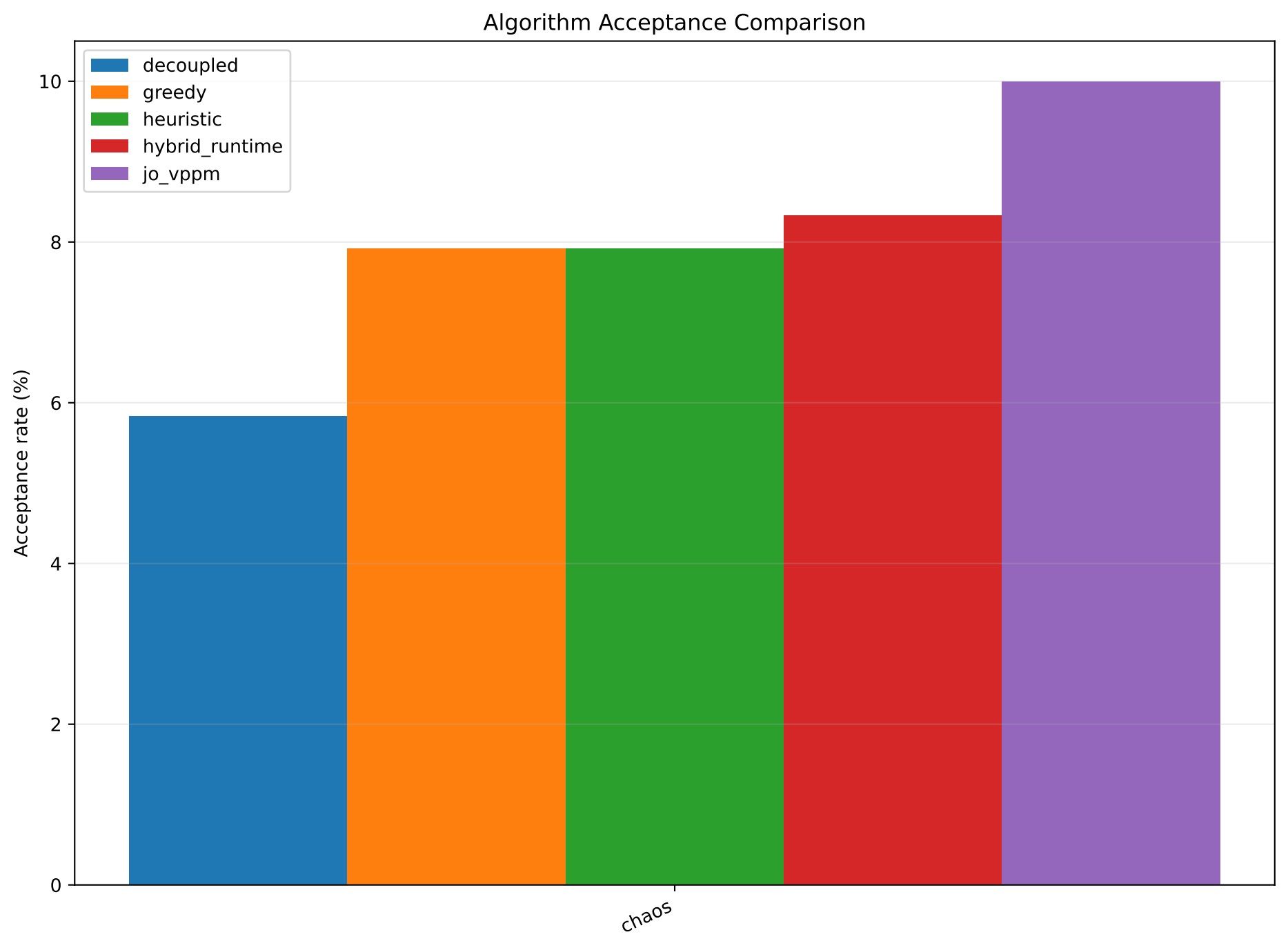
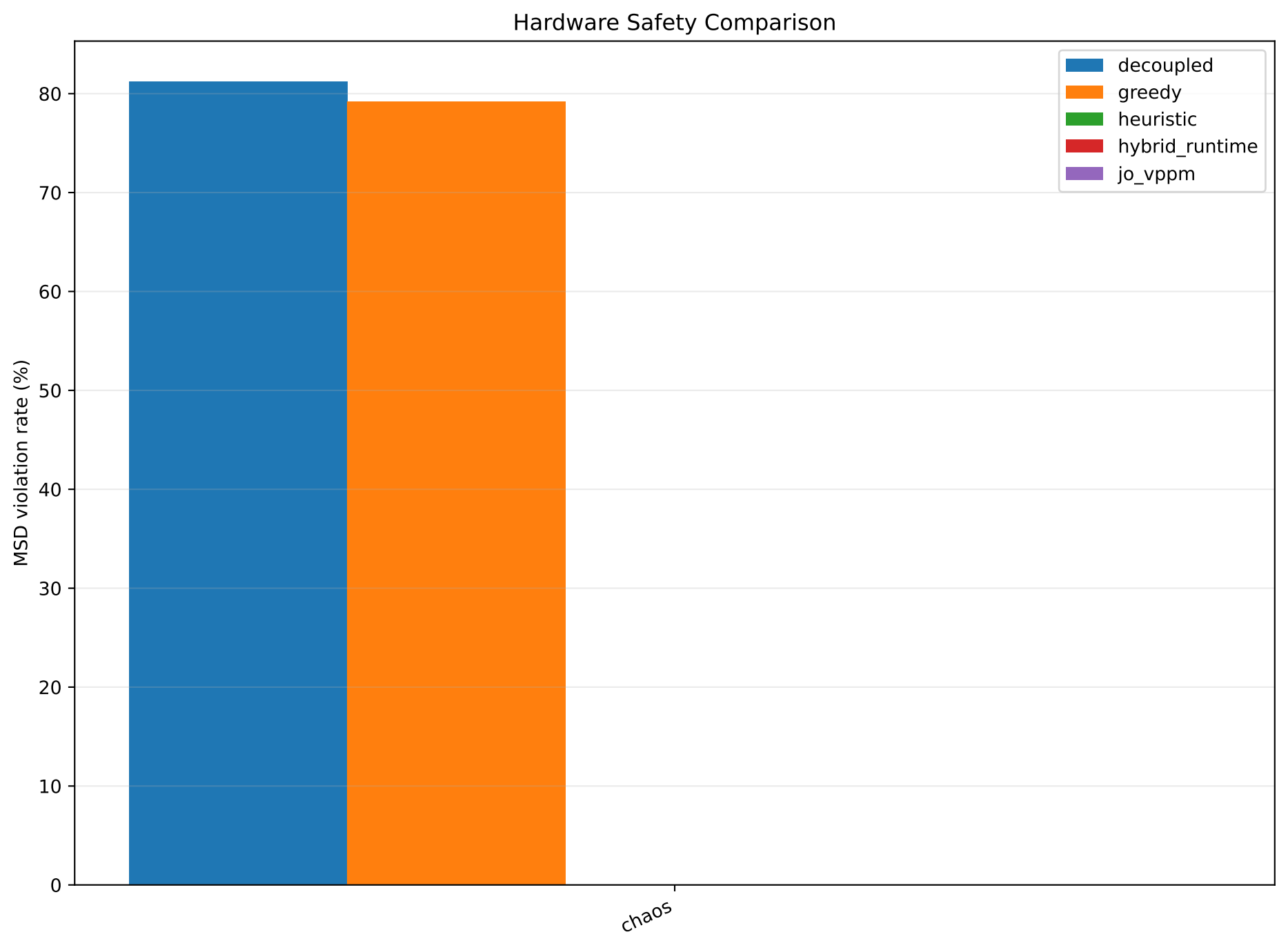
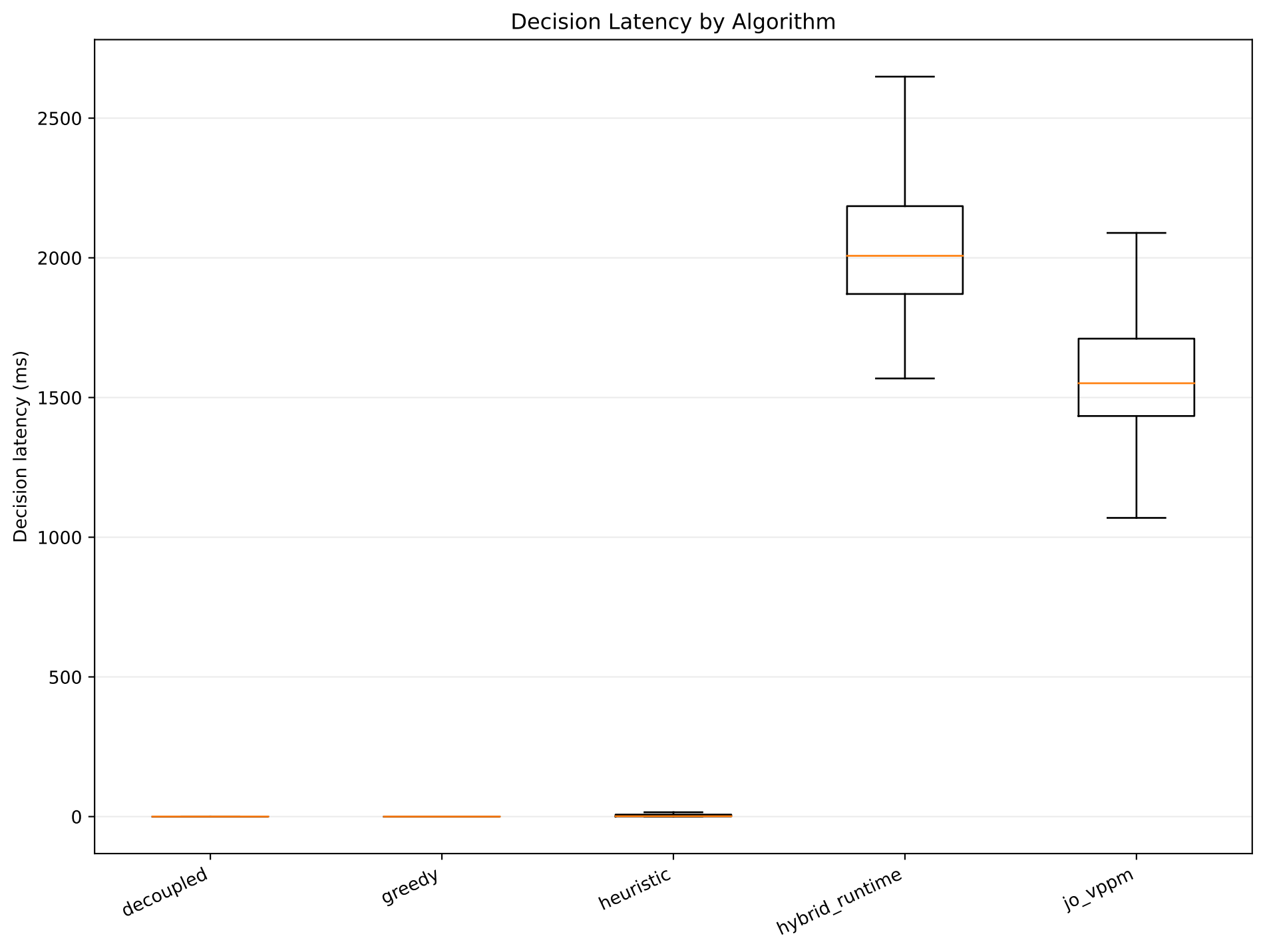
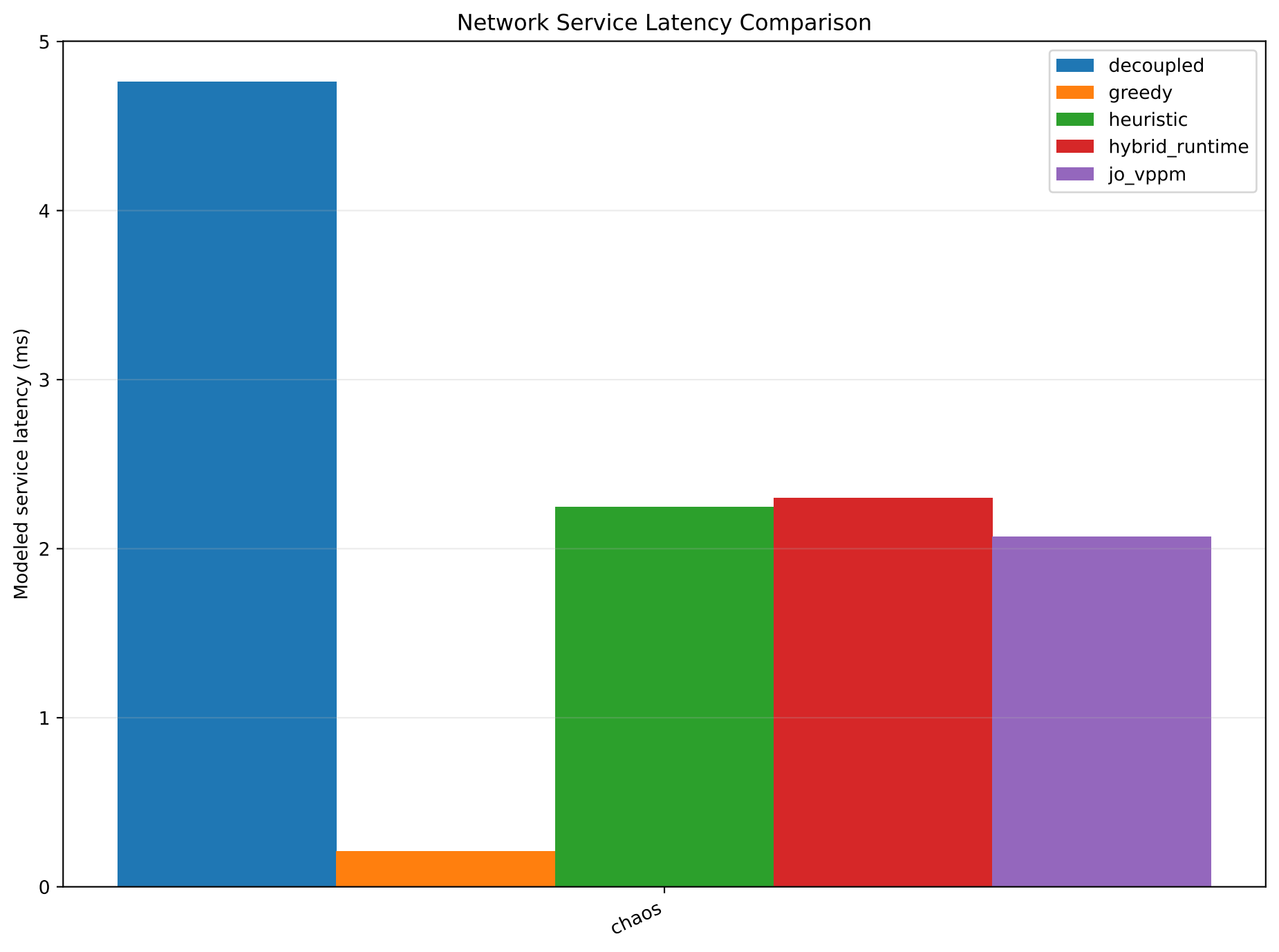
3S-COM Testbed Scenario Report - Scenario 5 - Chaos

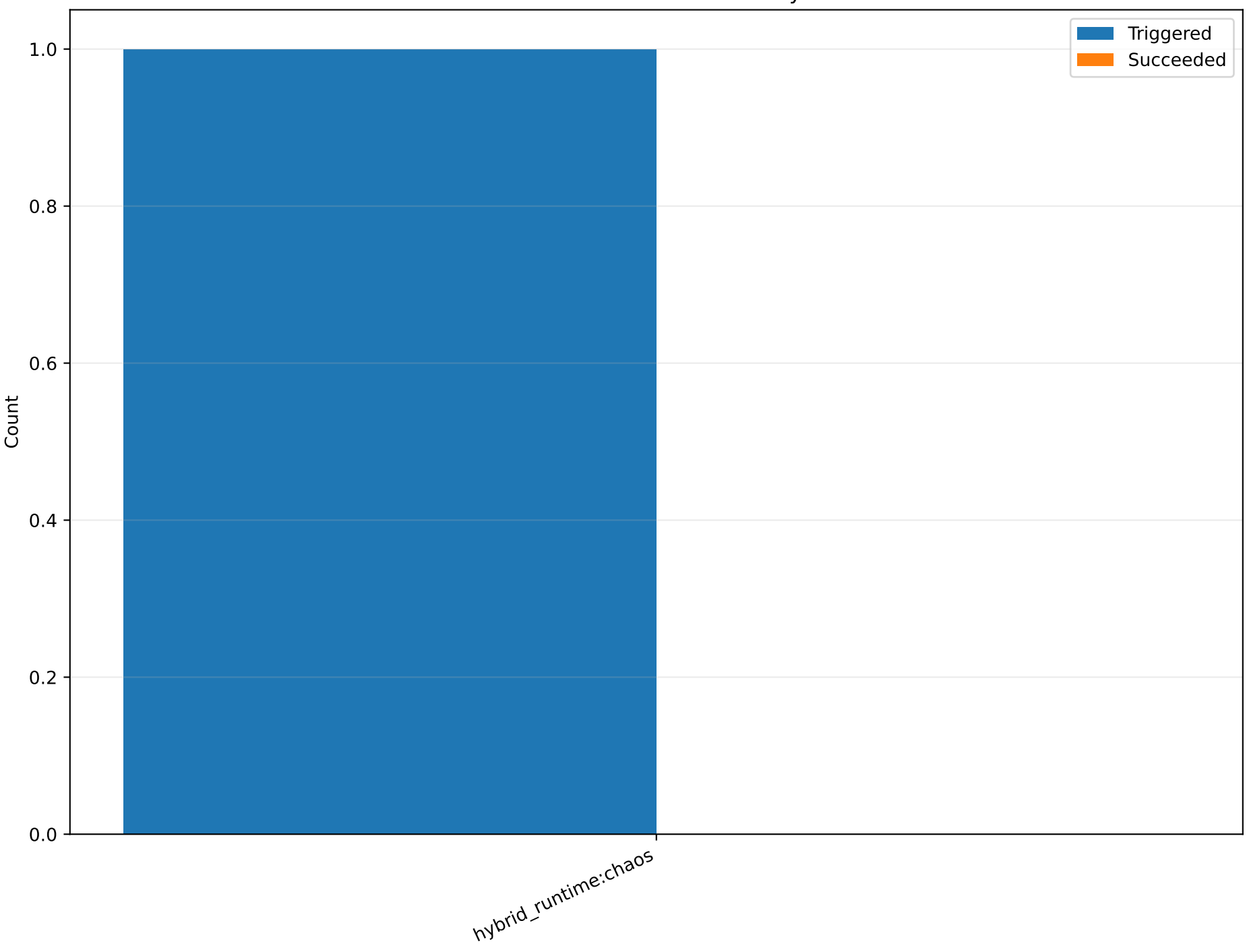
Scenario	Algorithm	Requests	Accept	No safe	MSD viol	Decision ms	Service ms	Timeouts	Migrations	MSD drops
Scenario 5 - Cha	hybrid_runtime	240	8.3%	91.7%	0.0%	2180.9	2.30	0	1	0
Scenario 5 - Cha	jo_vppm	240	10.0%	90.0%	0.0%	1640.9	2.07	0	0	0
Scenario 5 - Cha	heuristic	240	7.9%	92.1%	0.0%	9.0	2.25	0	0	0
Scenario 5 - Cha	greedy	240	7.9%	12.9%	79.2%	1.3	0.21	0	0	0
Scenario 5 - Cha	decoupled	240	5.8%	0.0%	81.2%	1.8	4.76	0	0	0


Algorithm Acceptance Comparison
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
Acceptance rate (%)
0
2
4
6
8
10
chaos


Hardware Safety Comparison
MSD violation rate (%)
0
10
20
30
40
50
60
70
80
chaos
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Decision Latency by Algorithm
Decision latency (ms)
2500
2000
1500
1000
500
0
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Network Service Latency Comparison
Modeled service latency (ms)
0
1
2
3
4
5
chaos
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Make-Before-Break Activity
Triggered
Succeeded
Count
1.0
0.8
0.6
0.4
0.2
0.0
hybrid_runtime:chaos